Chapter III.10

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II.8.6. *The Infinitesimal Lifting Property.* The following is very important in studying deformations of nonsingular varieties. Let k be an algebraically closed field, let A be a finitely generated k-algebra such that $\operatorname{Spec} A$ is a nonsingular variety over k. Let $0 \to I \to B' \to B \to 0$ be an exact sequence, where B' is a k-algebra, and I is an ideal with $I^2 = 0$. Finally, suppose we are given a k-algebra homomorphism $f : A \to B$. Then there exists a k-algebra homomorphism $g : A \to B'$ making the following diagram commute:

$$\begin{array}{ccc} & & 0 \\ & & \downarrow \\ & & I \\ & & \downarrow \\ & & B' \\ & \overset{g}{\nearrow} & \downarrow \\ A & \xrightarrow[f]{} & B \\ & & \downarrow \\ & & 0 \end{array}$$

We call this result the *infinitesimal lifting property* for A. We prove this result in several steps.

(a) First suppose that $g : A \to B'$ is a given homomorphism lifting f. If $g' : A \to B'$ is another such homomorphism, show that $\theta = g - g'$ is a k-derivation of A into I, which we can consider as an element of $\operatorname{Hom}_A(\Omega_{A/k}, I)$. Note that since $I^2 = 0$, I has a natural structure of a B-module and hence also of an A-module. Conversely, for any $\theta \in \operatorname{Hom}_A(\Omega_{A/k}, I)$, $g' = g + \theta$ is another homomorphism lifting f. (For this step, you do not need the hypothesis about $\operatorname{Spec} A$ being nonsingular.)

(b) Now let $P = k[x_1, \ldots, x_n]$ be a polynomial ring over k of which A is a quotient, and let J be the kernel. Show that there exists a homomorphism $h : P \to B'$ making the following diagram commute,

$$\begin{array}{ccc} 0 & & 0 \\ \downarrow & & \downarrow \\ J & & I \\ \downarrow & & \downarrow \\ P & \xrightarrow{h} & B' \\ \downarrow & & \downarrow \\ A & \xrightarrow{f} & B \\ \downarrow & & \downarrow \\ 0 & & 0 \end{array}$$

and show that h induces an A-linear map $\overline{h} : J/J^2 \to I$.

Proof. (a) Exactness gives us our map $\theta : A \to I$. Clearly θ is additive, and $\theta(1) = 0$ since $g(1) = 1 = g'(1)$, so that $\theta(k) = k \cdot \theta(1) = 0$. If $a, a' \in A$, then we have

$$\begin{aligned} a \cdot \theta(a') + a' \cdot \theta(a) &= f(a)\theta(a') + f(a')\theta(a) \\ &= g(a)\theta(a') + g'(a')\theta(a) \\ &= g(a)(g(a') - g'(a')) + g'(a')(g(a) - g'(a')) \\ &= g(aa') - g(a)g'(a') + g'(a')g(a) - g'(aa') \\ &= \theta(aa'), \end{aligned}$$

as desired.

Now, take $\theta \in \mathrm{Hom}_A(\Omega_{A/k}, I)$ and let $g' = g + \theta$. Clearly $g' : A \to B'$ is a k-module homorphism lifting f. If $a, a' \in A$, then

$$\begin{aligned} g'(a)g'(a') &= (g(a) + \theta(a))(g(a') + \theta(a')) \\ &= g(a)g(a') + g(a)\theta(a') + g(a')\theta(a) + \theta(a)\theta(a') \\ &= g(aa') + \theta(aa') \\ &= g'(aa') \end{aligned}$$

gives us that $g' : A \to B'$ is in fact a k-algebra morphism, as desired. □

10.1 Over a nonperfect field, smooth and regular are not equivalent. For example, let k_0 be a field of characteristic $p > 0$, let $k = k_0(t)$, and let $X \subset \mathbb{A}^2_k$ be the curve defined by $y^2 = x^p - t$. Show that every local ring of X is a regular local ring, but X is not smooth over k.

Proof. We begin by showing X is not smooth, and proceed to prove that each local ring at a closed point of X is regular, whence it will follow that all of its local rings are regular by (II, 8.14A).

Write $A = k[x, y]/(y^2 - x^p + 1)$, so that $X = \operatorname{Spec} A$. Any maximal ideal of $\mathfrak{m} \trianglelefteqslant A$ corresponds to a maximal ideal $\widetilde{\mathfrak{m}} \trianglelefteqslant k[x, y]$. Taking an algebraic closure $\overline{k}/k$, we have that the induced map $k[x, y] \hookrightarrow \overline{k}[x, y]$ is integral, so (AM, 5.10) gives us a maximal ideal $\mathfrak{n} \trianglelefteqslant \overline{k}[x, y]$ such that $\mathfrak{n} \cap k[x, y] = \widetilde{\mathfrak{m}}$.

Hilbert's Nullstellensatz says that $\mathfrak{n} = (x - a, y - b)$ for $a, b \in \overline{k}$. Then $q = \mathfrak{m} = (x - a, y - b) \cap A$ for $b^2 - a^p + t = 0$, and $A/\mathfrak{m} \cong k(a, b) \subset \overline{k}$. If $b = 0$, then $a = \sqrt[p]{t}$, so $k(\mathfrak{m}) = k(\sqrt[p]{t})$ has $k(\mathfrak{m})/k$ a non-separable extension. It follows from (II, 8.6A) that $\dim_{k(\mathfrak{m})} \Omega_{k(\mathfrak{m})/k} > 0$, For a closed point $q \in X$, stability under base exchange and (II, 8.4A) give us a surjection

$$\Omega_{X/k} \otimes k(q) \cong \Omega_{\mathcal{O}_q/k} \otimes k(q) \twoheadrightarrow \Omega_{k(q)/k}$$

with kernel $\mathfrak{m}_q/\mathfrak{m}_q^2$, In particular, we have $\Omega_{X/k} \otimes k(q) \neq 0$.

If $b \neq 0$, then $b = \sqrt{a^p - t}$, and $k(q)/k$ is separable, so $\mathfrak{m}_q/\mathfrak{m}_q^2 \to \Omega_{X/k} \otimes k(q)$ is surjective. If $a = \sum_i a_i t^i \in k$, then $b = \sqrt{\sum_i a_i^p t^{ip} - t} \notin k$ for degree reasons. □

10.2 Let $f : X \to Y$ be a proper, flat morphism of varieties over k. Suppose for some point $y \in Y$ that the fibre X_y is smooth over $k(y)$. Then show that there is an open neighborhood U of y in Y such that $f : f^{-1}(U) \to U$ is smooth.

Proof. None of this is even characteristic 0!!! □

10.3 A morphism $f : X \to Y$ of schemes of finite type over k is *étale* if it is smooth of relative dimension 0. It is *unramified* if for every $x \in X$, letting $y = f(x)$, we have $\mathfrak{m}_y \cdot \mathcal{O}_x = \mathfrak{m}_x$, and $k(x)$ is a separable algebraic extension of $k(y)$. Show that the following conditions are equivalent:
(i) f is *étale*;
(ii) f flat, and $\Omega_{X/Y} = 0$;
(iii) f is flat and unramified.

Proof. (i) $\Longrightarrow$ (ii): Let $i : X_y \hookrightarrow X$ be the inclusion. (II, 8.10) gives us $\Omega_{X_y/k(y)} \cong i^*(\Omega_{X/Y})$. i^* is a left adjoint and thus commutes with stalks, so

$$\begin{aligned}(\Omega_{X_y/k(y)})_x &\cong (\Omega_{X/Y})_x \otimes_{\mathcal{O}_x} (\mathcal{O}_X \otimes_{\mathcal{O}_y} k(y)) \\ &\cong (\Omega_{X/Y})_x \otimes_{\mathcal{O}_y} k(y).\end{aligned}$$

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$$\begin{aligned}(\Omega_{X/Y})_x &\cong (\Omega_{X_y/k(y)})_x \\ &\cong \Omega_{k(x)/k(y)}\end{aligned}$$

(ii) $\Longrightarrow$ (iii): It follows from (II, 8.10) and (II, 8.4A) that

$$\begin{aligned}0 &= \Omega_{X/Y} \otimes k(x) \\ &\cong \Omega_{X_y/k(y)} \otimes k(x) \\ &\cong \Omega_{\mathcal{O}_x/k(y)} \otimes k(x)\end{aligned}$$

maps surjectively onto $\Omega_{k(x)/k(y)}$. Then $\Omega_{k(x)/k(y)} = 0$, so it follows from (II, 8.6A) that $k(x)$ is a separable algebraic extension of $k(y)$. Now, we have a map $f^\sharp : \mathcal{O}_y \to \mathcal{O}_x$, and we wish to show that $\mathfrak{m}_y^e = \mathfrak{m}_x$. Clearly

(iii) $\Longrightarrow$ (i): f □

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10.9 Let $f : X \to Y$ be a morphism of varieties over k. Assume that Y is regular, X is Cohen-Macauley, and that every fibre of f has dimension equal to $\dim X - \dim Y$. Then f is flat.

Proof. It suffices, by localization of flatness, to show that $\mathcal{O}_x$ is flat over $\mathcal{O}_y$ for every closed point $x \in X$, where $y = f(x)$. Let $t_1, \ldots, t_r$ be a regular system of parameters for $\mathcal{O}_y$ □